

Connectivity of minimal Cayley graphs

By

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1. Introduction. Let G be a group with identity e . Let C be a subset of G such that $e \notin C$. The *Cayley graph* $X = X(G, C)$ has vertex set $V(X) = G$, where vertices g and h are adjacent if gh^{-1} or $hg^{-1} \in C$.

We call C a *minimal generating set* for G if C generates G but no proper subset of C does. Clearly a minimal generating set does not contain e . If C is a minimal generating set for G we say that $X(G, C)$ is *minimal*. We remark that $X(G, C)$ is connected if and only if C generates G . Hence a minimal Cayley graph is always connected.

Let X be a graph with vertex set $V(X)$. Assume X is not complete. Then $k_f = k_f(X)$ is defined to be the minimum number of vertices in a subset S of $V(X)$ such that $X \setminus S$ is disconnected with at least one finite component. If $V(X)$ is finite then $k_f(X)$ is nothing but the ordinary connectivity of X . For more information we refer the reader to [2].

Suppose now that $\text{Aut}(X)$, the automorphism group of X , acts transitively on the vertices of X . Let $d = d(X)$ be the degree of a vertex in X (we assume d is finite). Then it follows from the results in [2] that $k_f \geq (d+1)/2$ in general and, from [3, 4], that $k_f \geq 2(d+1)/3$ when X is finite. These lower bounds for k_f are achieved only if X is a lexicographic product of a finite complete graph by a two-way infinite path or a finite circuit. Less is known as to when k_f achieves the obvious upper bound, i.e., when is $k_f = d$? It follows from the results of [4] that this bound is attained when $\text{Aut}(X)$ acts primitively or edge-transitively on X and in certain other special cases [1].

In this note we show that if X is a minimal Cayley graph with finite degree d then $k_f = d$, thereby answering a question posed privately to the author by M. E. Watkins.

The problem of characterizing the Cayley graphs such that $d = k_f$ remains open, interesting and difficult.

2. Preliminaries. If $C \subset G$ we define

$$C^{-1} = \{c \in G : c^{-1} \in C\}.$$

From our definition of a Cayley graph we find that if $D = C \cup C^{-1}$ (where $C \subset G \setminus \{e\}$) then $X(G, C) = X(G, D)$.

Let X be a graph. If $S \subset V(X)$ then $N(S)$ denotes the set of vertices in $V(X) \setminus S$ that are adjacent to an element of S . Note that $k_f(X)$ is just the minimum value of $|N(S)|$ when S ranges over the finite subsets S of X such that $S \cup N(S) \neq V(X)$. If S is such a subset with $|N(S)| = k_f$ then it is said to be a *fragment* of X . A fragment S such that $|S|$ is a minimum is an *atom*. (Warning: our definition of the term "fragment" differs slightly from that in [2].)

2.1. Lemma. *Let X be a graph such that $A = \text{Aut}(X)$ acts transitively on the vertices of X and let d be the degree of a vertex in X . Suppose that $k_f(X) < d$. Then $V(X)$ has a partition into sets $Y_1, \dots, Y_m$ such that:*

- (a) *the Y_i form a complete block system for $\text{Aut}(X)$,*
- (b) *the subgraph induced by each Y_i is connected, with at least two vertices,*
- (c) *$N(Y_i)$ is the union of those Y_j which contain a vertex adjacent to a vertex in Y_i . (Hence $|N(Y_i)| = k_f(X)$).*

Proof. Let Y be an atom of X . If $g \in \text{Aut}(X)$ then Y_g , the image of Y under the action of g , is also an atom. By [2: Corollary 1A], distinct atoms of X are disjoint. Accordingly the set of all translates Y_g of Y ($g \in \text{Aut}(X)$) forms a complete block system for $\text{Aut}(X)$. We denote the elements of this set by $Y_1, \dots, Y_m$ where $Y = Y_1$.

Since $\text{Aut}(X)$ acts transitively on the set of the Y_i , we may complete the proof of the lemma by verifying that (b) and (c) hold for Y .

Let V_1 induce a component of Y . Since $V_1 \subseteq Y$ we see that $N(V_1) \subseteq N(Y)$. As Y is a fragment this implies that $N(V_1) = N(Y)$ and so V_1 is a fragment. Since Y is an atom $|V_1| \geq |Y|$, so we conclude that $Y = V_1$ and hence that Y is connected. If Y consisted of a single vertex then we would have $|N(Y)| = d > k_f$, a contradiction. Thus $|Y| \geq 2$ and (b) is proved.

Using Corollary 1A of [2] again, we know that either $Y_i \subseteq N(Y)$ or $Y_j \cap N(Y) = \emptyset$. Since the Y_i partition $V(X)$ this implies that $N(Y)$ is a union of some of the Y_j . This yields (c). $\square$

3. The main result.

3.1. Theorem. *Let C be a finite minimal generating set for the group G and let $X = X(G, C)$. Then $k_f(X) = |C \cup C^{-1}|$.*

Proof. Let $d = |C \cup C^{-1}|$. We assume by way of contradiction that $k_f(X) < d$. Let $Y_1, \dots, Y_m$ be the partition of $V(X)$ provided by Lemma 2.1. Set $Y = Y_1$ and assume $e \in Y$.

We recall that since X is a Cayley graph for G , we may assume that G acts regularly on X . Since the Y_i form a block system for $\text{Aut}(X)$, they also form a block system for G .

Let H denote the subgroup of G fixing Y , as a set. We claim Y is a minimal Cayley graph for H . Since Y is a block for G we see that H acts transitively on Y . As G acts regularly on X it follows that H acts regularly on Y .

Let D_1 denote the set of vertices in Y adjacent to e . Using the fact that H acts regularly on Y it is straightforward to verify that $Y = Y(H, D_1)$. Let $D = D_1 \cap C$. Since $D_1 \subset C \cup C^{-1}$ we find that $D_1 = D \cup D^{-1}$ and so $Y = Y(H, D)$. As Y is connected (by 2.2(b)), D must generate H .

On the other hand $C = (C \setminus H) \cup D$ is a minimal generating set for G and therefore, D must be a minimal generating set for H . Thus our claim is established.

We wish now to count the number of the Y_i which contain a neighbour of e , and thereby get a lower bound on $|N(Y)|$. It is easy to verify that $V(Y_i)$ is a right coset of $H = V(Y)$. Therefore Y_i contains a neighbour of e iff $V(Y_i) = Hx$ for some x in C .

Suppose x, y are distinct elements of $C \cup C^{-1}$ such that $x \notin H$ and $Hx = Hy$. Then $xy^{-1} \in H$, which implies that x lies in the subgroup of G generated by $D \cup \{y\}$. Since C is minimal and $x \notin H$ we must therefore have $x = y^{-1}$. As $xy^{-1} \in H$ this means that $x^2 \in H$ and as $x \neq y$, $x^2 \neq e$.

Let $c = |C \cup C^{-1}|$, $d = |D \cup D^{-1}|$. If we set

$$\begin{aligned} u &= |\{x \in C \cup C^{-1} \mid x \notin H, e \neq x^2 \in H\}|, \\ v &= (c - d) - u \end{aligned}$$

then the number of distinct cosets $Hx \neq H$ containing a neighbour of e is $\frac{1}{2}u + v$. By Lemma 2.1(c) each of these cosets lie in $N(Y)$. We thus have

$$(1) \quad (\tfrac{1}{2}u + v)m \leq |N(Y)| < c = u + v + d.$$

Assume first that $u = 0$. Since $e \notin D$, $d + 1 \leq m$ and so (1) yields $v(d + 1) < d + v$, whence $v = 0$. But if $u = v = 0$ then $c = d$ and so $G = H$, which is impossible. We assume then that $u \neq 0$. (Consequently $u \geq 2$, since by its definition it is even.) Then there exists $x \in C \cup C^{-1}$ such that $x \notin H$ and $x^2 \in H \setminus e$. Since C and D are minimal x^2 cannot lie in $D \cup D^{-1}$ and therefore we must have $d + 2 \leq m$. From (1) we now find that

$$(\tfrac{1}{2}u + v)(d + 2) < d + u + v$$

and therefore

$$(\tfrac{1}{2}u - 1)d + v + vd < 0.$$

As $u \geq 2$ this is impossible. $\square$

Acknowledgement: I would like to thank the referee, whose suggestions simplified and shortened my original proof of Theorem 3.1.

References

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Eingegangen am 31. 10. 1980 *)

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*) Eine Neufassung ging am 12. 5. 1981 ein.